

Dongha Lee

Stanford, CA, USA | Seoul, Republic of Korea

dhalee@stanford.edu | jundong9827@gmail.com | +82 10-2746-0818

Education

Stanford University

M.S. in Electrical Engineering

Stanford, CA, USA

Sep. 2026 – Expected June 2028

- Incoming M.S. student, Autumn Quarter 2026.
- **Research Interests:** Foundation models, parameter-efficient fine-tuning, representation learning, and information-theoretic approaches to deep learning.

Chung-Ang University

B.S. in Electrical and Electronics Engineering

Seoul, Republic of Korea

Mar. 2019 – Feb. 2026

- **GPA:** 4.40 / 4.50 (**99.00** / 100) | *Summa Cum Laude* | **Rank:** 8/159

Publications

- [1] **Dongha Lee***, Jinhee Park*, Minjun Kim, and Junseok Kwon, “ABM-LoRA: Activation Boundary Matching for Fast Convergence in Low-Rank Adaptation.” *Submitted to the AAAI Conference on Artificial Intelligence (AAAI)*, 2027. [Under Review] [[arXiv](#)] [[Code](#)]
*Equal contribution
- [2] **Dongha Lee** and Junseok Kwon, “Designing Information Flow in Deep Representations: The Importance of Information Bottleneck and a Case Study in Invariant Learning.” In *Proceedings of the Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Jeju, Korea, 2025. (Poster Presentation) [[Supplementary Material](#)]

Research Experience

Computer Vision & Machine Learning (CVML) Lab

Undergraduate Research Intern | Advisor: Prof. Junseok Kwon

Chung-Ang University

Nov. 2024 – Aug. 2026

ABM-LoRA — Activation Boundary Matching for Low-Rank Adaptation

Under Review at AAAI 2027, co-first author

- Found that activation boundaries stabilize earlier than activation values during adaptation, and distilled them from a short probe adapter to initialize LoRA with no inference overhead.
- Benchmarked on ConvNeXt-T, Swin-T, ViT-B/16, T5-base, and LLaMA2-7B against PiSSA, OLoRA, EVA, and LoRA-One.

Information-Theoretic Representation Learning (IB/IIB)

IEIE Summer Conference 2025 — First author, Poster Presentation

- Consolidated IB/IIB theoretical foundations to characterize information retention and disentangle causal vs. spurious features in deep representations.
- Insights from this analysis — particularly on which representational structures carry task-relevant information — motivated the boundary-based transfer strategy behind ABM-LoRA.

HI-LoRA: Hybrid Initialization for Low-Rank Adaptation [[Code](#)]

Undergraduate Thesis (Independent Research)

Chung-Ang University

June 2025 – Dec. 2025

- Designed HI-LoRA, a hybrid initialization combining gradient-based (LoRA-GA) and magnitude-based (EVA-LoRA) methods to preserve both directional and scale information.
- Improved early-training stability and lowered seed variance, indicating that LoRA initialization errors stem from both magnitude distortion and tangent-space misalignment.

ESG Practical Problem-Solving Research Program

Team Leader & Lead Researcher | Advisor: Prof. Jung-Ryun Lee

Chung-Ang University

May 2024 – Dec. 2024

- Led a 5-person interdisciplinary team, setting technical direction and coordinating across engineering backgrounds.
- Built a noise-robust hybrid recognition system combining LipReading and ASR to counter performance degradation in high-noise environments.
- Implemented a PSNR-based adaptive switching algorithm balancing visual–audio modalities to maintain recognition accuracy where standard ASR fails.

Awards & Honors

- **Summa Cum Laude**, Chung-Ang University (Feb. 2026)
Highest graduation honors, School of Electrical and Electronics Engineering.
- **Academic Excellence Scholarship**, Chung-Ang University
Awarded for outstanding academic performance over 5 semesters (Fall 2025, Spring 2025, Fall 2024, Spring 2020, Fall 2019).
- **Alumni Scholarship**, Chung-Ang University (Dec. 2019)

Skills

- **Languages:** Python (primary), C/C++, MATLAB, Bash
- **Deep Learning:** PyTorch, HuggingFace (Transformers, PEFT, Accelerate), NumPy, Pandas
- **Infrastructure:** Multi-GPU training (4×RTX 3090), Linux/Ubuntu, SSH, Conda, Git, CUDA

Leadership & Project Management

CUAI (Artificial Intelligence Club)

Mar. 2025 – Jul. 2025

Research Member

- Delivered a technical presentation on Google’s Titan foundation models, analyzing architectural design and performance trade-offs as part of weekly research seminars.

Da Vinci Volunteer Group

Mar. 2019 – Dec. 2020

President

- Directed a virtual mentoring program during the pandemic, overseeing curriculum planning and communication with local institutions.
- Led outreach initiatives at rural high schools, mentoring 36 underserved students on academic and career pathways.

Creative & Professional Experience

Fashion Photographer & Assistant [\[Portfolio\]](#) [\[Instagram\]](#)

Mar. 2023 – Jan. 2024

- Assisted lead photographers in commercial studio workflows for major clients (e.g., Netflix, Kakao), supporting corporate and editorial projects.
- Managed end-to-end technical workflows at Grant Studio, including site surveys, lighting setup, equipment calibration, and on-site logistics.
- Covered the 2023 Australian Fashion Week (AAFW) as an accredited photographer.

Finders Family (Photography Club)

Nov. 2019 – Nov. 2020

Exhibition Coordinator

- Organized the club’s first virtual exhibition, managing the curation and online hosting platform.
- Led photo critique and editing sessions, mentoring members on visual composition and post-processing.